

Research article

A convolutional autoencoder framework for ECG signal analysis

Ugo Lomoio ^{a, , 1}, Patrizia Vizza ^{a, 1}, Raffaele Giancotti ^a, Salvatore Petrolo ^b, Sergio Flesca ^b, Fabiola Boccuto ^d, Pietro Hiram Guzzi ^{a, , *}, Pierangelo Veltri ^{b, }, Giuseppe Tradigo ^c

^a Department of Surgical and Medical Sciences, University of Catanzaro, Catanzaro, Italy

^b DIMES, University of Calabria, Rende, Italy

^c SMARTTEST Laboratory, E-Campus University, Novedrate, Italy

^d Division of Cardiology, Department of Surgical and Medical Sciences, University of Catanzaro, Catanzaro, Italy



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ABSTRACT

Electrocardiographic (ECG) signals are used to evaluate heart activity and to identify disease-related anomalies. Reliable support systems are useful for analyzing ECG signals, for instance, in long-term data acquisition and evaluation (e.g., 24-hour holter recording) or to support physicians in reading ECGs. Analysis of time varying signals may be done by using autoencoders (AEs) deep neural networks. AE specialized for signal data, named Convolutional Autoencoder (CAE), showed the best performances in the analysis of ECG signals.

This paper presents a CAE-based framework for ECG signal analysis and anomaly identification. The trained phase is performed on synthetic data signals. The trained neural network obtained is used for the detection of anomalies in ECG signals. The trained framework has been tested on 12 lead ECG signals on a benchmark dataset and applied in scenarios where anomalies are related to cardiological risks and pathologies. The results show interesting results in automatic anomaly detection to support physicians in the decision process. The results show that the CAE-based framework is able to identify anomalies in ECG signals with a ROC AUC of 97.82% on simulated test set and a ROC AUC of 99.75% using a real test set. Finally, the proposed method has been enriched by means of reconstruction error based explainability modules and time-windows based preprocessing modules. Explainability results have been validated using abnormalities annotated by a cardiologist as ground truth and compared with explanations results. System with both code and data, is available at https://github.com/UgoLomoio/ECG_DSS_CAE.

1. Introduction

ECG signals are commonly used to assess heart activity and health status, particularly in evaluating the risk of heart attack [1]. ECGs are sets of signals that measure the electrical activity of the heart. The duration of an ECG signal can range from 10 seconds, as used in standard clinical evaluations, to several days in the case of continuous cardiac monitoring (holter device) [2]. Heart activity is typically measured through multiple channels, which can range from 1 to 12. Some ECG devices are also capable of measuring

* Corresponding author.

E-mail address: hguzzi@unicz.it (P.H. Guzzi).

¹ These authors equally contributed.

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three additional signals to generate vectorcardiograms (VCG), which trace both the magnitude and direction of the electrical forces associated with heart activity. Wearable devices now allow real-time measurement and monitoring of heart-related signals even during physical activities.

Although understanding ECG signals is a primary concern for physicians, there remains a need for reliable tools to interpret and analyze these signals automatically. For example, cardiac arrhythmias are characterized by irregular heartbeats, and detecting them involves identifying deviations from a regular rhythm. By reading electrocardiograms, clinicians can detect conditions such as atrial fibrillation and frequency anomalies [3,4]. Similarly, arrhythmias can be identified by experts through a visual examination of ECGs [5,6]. In long-term ECG signal acquisition, computer-assisted analysis tools are essential for aiding physicians in the detection of anomalies. Consequently, automated anomaly detection in ECG signals has emerged as a primary research focus. In this direction, AI-based tools can assist clinicians by integrating the insights gained from studies and analyses [7–14].

This work explores the use of Convolutional Autoencoder (CAE) [15], a neural network architecture that has recently gained traction in signal analysis research. Compared to standard Autoencoders (AEs), CAEs are more effective in capturing spatial patterns and relationships in ECG data, making them particularly suited for anomaly detection. Moreover, we propose a new module for preprocessing and an explainable module to increase the efficiency and efficacy of the automatic detection of ECG anomalies. This direction is in the same direction of relevant studies such as in [16], where the use of long short-term memory (LSTM)-based autoencoder is proposed to differentiate between normal and atrial fibrillation (AF) ECG segments.

This paper proposes a CAE-based tool for the analysis of ECG signals and the detection of anomalies. The tool identifies abnormal windows in ECG signals, with experimental settings designed to test the system against atrial fibrillation, myocardial infarction, and noise disturbances in the signals. Tests have been conducted using both real and synthetic datasets, and the results have been compared to other ECG signal analysis tools. The experimental outcomes demonstrate high accuracy proving the reliability of the proposed tool in analyzing ECG signals. The number of detected anomalies was measured using the Missed Alarm Rate (MAR) [17] to quantify the accuracy of anomaly identification. The proposed tool achieved a MAR of 0.01%, indicating it can correctly identify nearly all anomalies. The proposed CAE solution is also enriched by means of an explainable process that supports clinicians in deciding how anomaly is better classifying the detected anomalies. By using the here proposed CAE, clinicians can classify anomalies in the signals in a more rapid way. For instance, clinicians can rapidly decide if the presented anomalous window is related to a particular cardiac disease (both beat and rhythm level) or is a healthy structural anomaly that can be present in the ECG. We tested such a methodology in the real case with the support of cardiologists of the University Magna Graecia Medical Hospital. We also proved explainability performance by comparing CAE locally identified anomalies, with patterns annotated by the expert cardiologist.

The application of the proposed CAE is relevant with respect to the importance of anomaly detection in ECG signals for improving healthcare outcomes. By identifying abnormal heart rhythm patterns, the proposed system can aid in the early diagnosis of cardiovascular diseases such as arrhythmias, myocardial infarction, and heart failure. In clinical environments, anomaly detection algorithms can be integrated into wearable devices or medical monitoring systems to provide real-time alerts, enabling timely intervention and potentially preventing adverse health events.

In hospital settings, a CAE-based anomaly detection framework can be integrated into patient monitoring systems to identify subtle but significant changes in ECG patterns, such as those indicating the onset of an acute cardiac event by using also wearable devices, allowing continuous ECG monitoring, enabling real-time anomalies detection (e.g., arrhythmias) generating alerts for medical attention. Additionally, anomaly detection in ECG signals can be used to analyze large ECG datasets to identify new disease markers or evaluate the efficacy of therapeutic interventions. By identifying patterns related to specific diseases or treatment responses, this system contributes to a deeper understanding of cardiovascular pathophysiology and enhances clinical decision-making. With respect to available autoencoder based solutions allowing anomaly detection on 12-leads ECG signals, the novelty of this work is in proposing a general anomaly detection framework, tested on different beat and rhythmic abnormalities, with an explainable module able to give clinicians more detailed insights on abnormalities location. Finally, the CAE-based framework can be used for signal reconstruction and feature extraction through signal seeding where a single seed can be employed as a query filter on ECG signals to identify known anomalies efficiently.

2. Related work

The automatic analysis of ECG signals focuses on extracting information about heart activity and identifying anomalies. A recent objective in signal analysis is the ability to automatically filter out anomalies that are not associated with clinical conditions. This is achieved through machine learning applications for anomaly detection and class association (i.e., identifying whether a condition is present). For example, Li et al. [18] discuss various machine learning and deep learning models designed to detect individual anomalous heartbeats in ECG signals. These methods were trained and evaluated using the MIT-BIH Arrhythmia Database, which includes 2-channel ECG recordings [24,19]. There exist examples of using AutoEncoders (AE) for anomaly detection in ECG signals. In Liu et al. [20], a Long Short Term Memory AE (LSTM-AE) model is used to perform ECG anomaly detection with an accuracy of 98% with a reconstructed error threshold of 0.02 in terms of absolute error loss. Similarly, Anomagram uses AE to perform an anomaly detection task [21]. It uses a Dense AE (DAE) to perform an ECG anomaly detection task, tested on the ECG5000 dataset [22], focusing on distinguishing if a given ECG window is normal or abnormal with an accuracy of 95%. An abnormal signal does not necessarily indicate the presence of a pathology. Hou et al. in [23] present an LSTM-AE with a Support Vector Machine (SVM) to classify five different types of heartbeats in the MIT-BIH ECG dataset [24]: normal beat, left bundle branch block, right bundle branch block, atrial premature complexes, and premature ventricular contractions. The proposed model reports an accuracy of 99.74%, a sensitivity of 99.35%, and a specificity of 99.84%. Results on real datasets are similar to the ones we performed on our tests, but on

a reduced number of ECG leads. Moreover, we propose an explainable module able to support clinicians in the decision process. We used local explainability method based on the *reconstruction error based method* [25] for methods based on signal reconstructions (e.g., GAN based). We considered the explanations using SHAP [26], [27] but the obtained explainable results are less reliable with respect to the ones obtained by using reconstruction error based method. [28] uses LSTM with autoencoder to identify anomalies in ECG using 2 channels. In Roy et al. [29] an LSTM-AE model used on ECG signals, achieved an accuracy of 98.42% with precision, recall, and F1 values of 94.52%, 98.22% and 96.33%, respectively. Shan et al. [30] proposed a framework to perform ECG anomaly detection based on an adversarial AE and temporal convolutional network (TCN) which consists of three modules: an AE, the discriminator, and an outlier detector. The framework showed good performance in the classification of ECG signals. [31] proposes the use of a GAN for anomaly detection on two channels of ECG signals. Similarly to our approach anomaly detection is performed by using reconstruction error with a user-defined threshold. The experimental results reach interesting performance (i.e., 0.954 ROC AUC), while the here proposed framework reaches better performance. However, the dataset used is different from the one here used (i.e., PTB dataset).

Even if AE has been used for analysis and physicians have supported studying ECG analysis, the here presented contribution still finds room in the literature. Most existing solutions focus on detecting specific beat-level anomalies (e.g., atrial fibrillation) in ECG signals acquired with fewer than 12 channels (e.g., the 2-lead MIT-BIH dataset). Our approach seeks to address the need for a more comprehensive anomaly detection framework capable of identifying various beat- and rhythm-level abnormalities. To achieve this, we apply a window-splitting method to preprocess the signal, utilizing different modules and methodologies based on time-window analysis. Also, we used a synthetic data set to train our AE-based convolutional framework, tuning the input dataset and reaching a more general-purpose trained model.

Authors in [16] use the PTB-XL [32] 12-lead ECG dataset to train an LSTM based autoencoder to identify only atrial fibrillation (AF) anomalies in ECG beats. Model was tested using a different 12-lead ECG dataset, the China Physiological Signal Challenge 2018 (CPSC 2018) [33]. To compare their performances against ours, we replicated one of their experiments regarding atrial fibrillation detection using only QRS complexes. The novelty of the here proposed framework regards the accuracy obtained by training our model using both synthetic and real healthy ECG data. We also differ in the input we provide to the model, by using windows instead of single-beats we can focus on both beat-level and rhythm-level anomalies. Also in [34] the PTB-XL dataset is used to test a multimodal network. Reached performances are 0.86 ROC AUC.

In [35] authors propose an explainable denoising autoencoder to perform anomaly detection on the CPSC 2021 dataset obtaining a ROC AUC of 0.883 but a Missed Alarm Rate of 50.46%. Using explainability techniques authors aim to highlight the anomalous beats present in the input window. In contrast, the here proposed framework introduces a local explainability method based on error reconstruction, aiming to give clinicians more detailed insights on abnormalities location (i.e., abnormal P wave).

Finally, the here presented CAE-based classification tool here is able to analyze large time interval ECG recorded signals, such as the ones recorded by 7 days holter. The comparison with available tools that perform anomaly detection in ECG signals indicates that this method performs better.

3. Materials and methods

Fig. 1 reports the general purpose architecture of an autoencoder (AE) [15], including Encoder and a Decoder Encoder modules. The encoder compresses the acquired data (i.e., ECG windows) and maps them in a latent space, reducing the data dimensions. The decoder is applied to latent space to map compressed representation back to the original domain (i.e., the input data). The AE model is mainly based on a neural network trained to compress and reconstruct input using a feedforward approach based on Mean Square Error (MSE) between the input signal and the reconstructed one returned as output. MSE measures the average squared difference between the actual and reconstructed signal as follows:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

where y_i represents the actual value, $\hat{y}_i$ the reconstructed value, and n the number of samples in the signal.

A Convolutional Autoencoder (CAE) extends a standard AE by incorporating convolutional layers. Fig. 2 reports an example of a convolutional layer used to extend an AE module. We used a CAE-based framework to improve feature learning and signal reconstruction, offering greater noise resistance, and thus preserving spatial information.

CAE performs very well in reconstructing signals or images, presenting noise robustness, and preserving spatial information. The presented CAE comprises two-dimensional convolutional layers for both the encoder and the decoder, while a one-dimensional convolutional layer makes the ending module (the OutputModule) in the decoder. This architecture allows us to leverage the benefits of processing and reconstructing the input signal. The MSE error between the reconstructed and original signals is then used to detect anomalies [36,37].

To train and test the CAE, both synthetic and real 12-lead ECG signal windows were used. The datasets were initially divided into normal and abnormal (i.e., containing anomalies) signal windows. 80% of the normal windows were used for model training, while the remaining 20% was combined with the abnormal dataset to create the test set for model evaluation. Given an ECG signal, the following preprocessing modules have been implemented and used: (i) a band-filter of 0.5-100 Hz to reject noise such as baseline wander and EMG; (ii) a notch filter to attenuate 50 Hz interference caused by the electrical grid; (iii) a re-sampling module applied to the filtered signals to change the sampling rate from 1000 Hz to 500 Hz; (iv) a windowing module to extract 4-second length windows from the ECG signal.

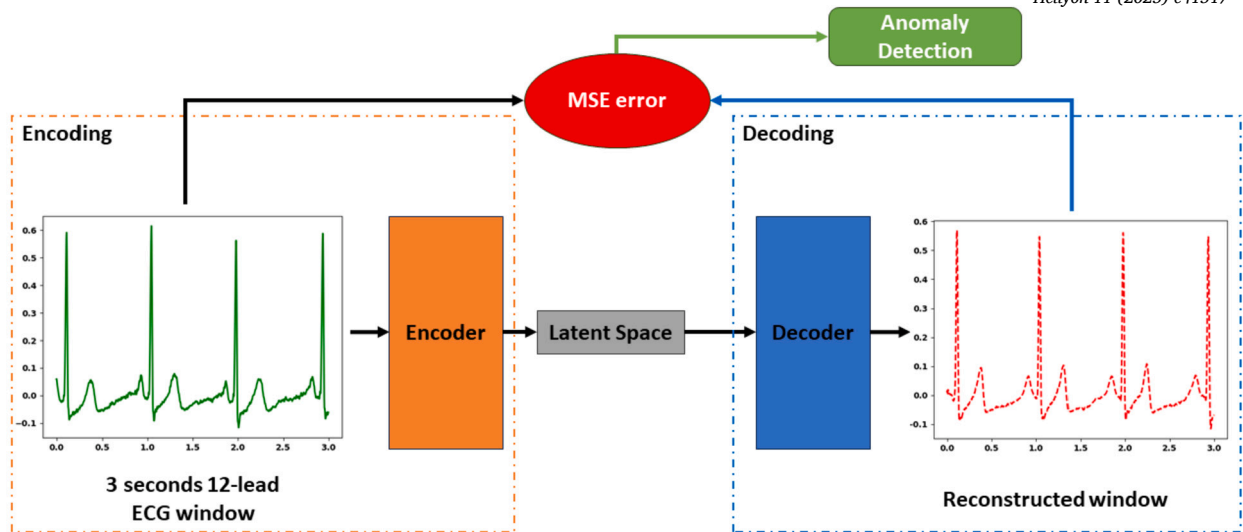


Fig. 1. General AE architecture with Encoder, Decoder, and Latent Space. Given an input signal, the encoder maps it into a latent space. Then, the decoder is responsible for reconstructing the input signal. The reconstruction error (MSE Error) is minimized during the training phase using signals belonging to the same class. The architecture is used as a discriminator under the hypothesis that signals belonging to a different class produce a higher error.

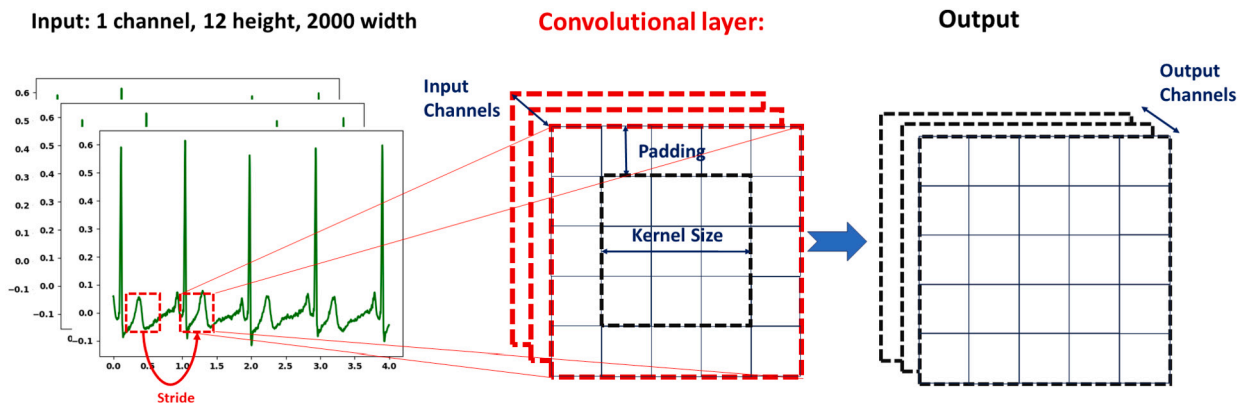


Fig. 2. Input, parameters, and output of a convolutional layer.

Synthetic ECG signals were generated with the SIMARR3 simulator [38]. The synthetic data generator tool allowed one to obtain almost 500 ECG signals with and without anomalies. The generated ECG signals contain anomalies that can be associated with pathologies (i.e., atrial fibrillation, tachycardia, bradycardia). Input parameters have been set to have no noise to the signal generator, which means that all signals can be considered ECG signals, normal (no pathologies), and with anomalies.

The training simulated set consists of 400 ECG signals without arrhythmia and noise (labeled as *normal*) sampled at 500 Hz and with different time duration. A simulated test set consisting of 100 standard ECG signals and 100 ECG signals with anomalies (labeled *anomaly*), has been used. Both were sampled at 500 Hz and with different time duration. As anomalies, we simulate cardiac anomalies such as atrial fibrillation events and arrhythmia. A windowing algorithm has been applied to both simulated train and test sets to extract four-second-long signal windows from each signal.

The database used for testing is the PTB Diagnostic ECG Database version 1.0 [39]. It has been used for retraining and testing the CAE-based framework. The PTB database contains a total of 549 signals from 290 different patients. Each record includes 15 simultaneously measured channels from two different signals:

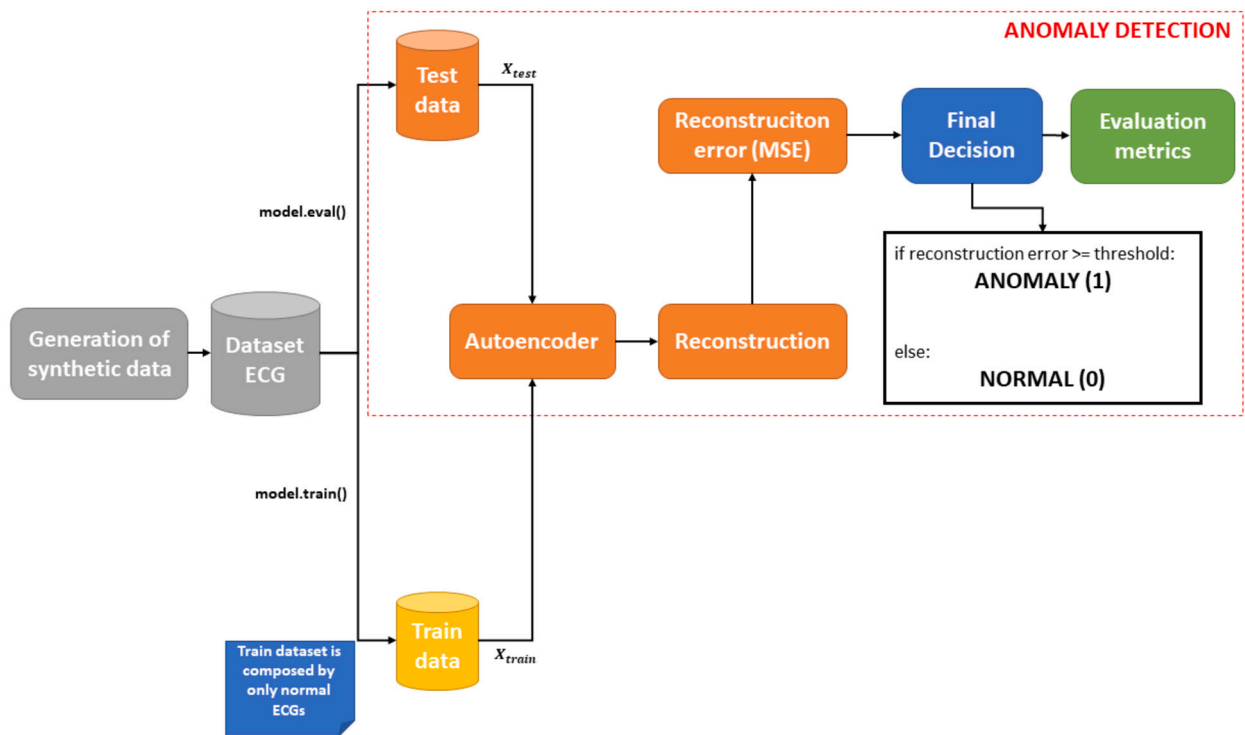
- the conventional 12 ECG leads (i, ii, iii, avr, avl, avf, v1, v2, v3, v4, v5, v6);
- the 3 leads VCG (x, y, z).

Each signal in the PTB dataset has been digitized with a sampling rate of 1000 Hz. A resampling phase is required to change the sampling rate from 1000 Hz to 500 Hz. A total number of 22 ECG signals of the PTB dataset were excluded because the diagnostic classification of the signal was unavailable. The diagnostic class distribution of the remaining 268 patients is summarised in Table 1. The diagnostic pathologies listed in Table 1 were all used for testing the model. The healthy controls were randomly split into training

Table 1

Data extracted from the PTB dataset used to test the CAE-based framework.

Diagnostic class	Number of patients	Number of ECG signals
Myocardial infarction	148	368
Cardiomyopathy/Heart failure	18	20
Bundle branch block	15	17
Dysrhythmia	14	16
Myocardial hypertrophy	7	7
Valvular heart disease	6	6
Myocarditis	4	4
Palpitation	1	1
Stable angina	1	1
Unstable angina	2	2
Healthy controls	52	80

**Fig. 3.** General workflow for anomaly detection using an AE model.

and test sets, with 80% allocated for model fine-tuning and 20% reserved for the test set. The extracted ground truth targets were mapped into normal/anomalous, to obtain: 52 normal patients (80 normal signals) and 216 anomalous patients (442 anomalous signals).

The CAE-based framework has been trained on standard 12-lead ECG signal windows, allowing for efficient reconstruction of standard ECG windows (with low MSE values) throughout the training process. The model was able to reconstruct ECG signal windows both with and without anomalies. Differences in reconstruction between normal and abnormal ECG windows can be used to classify whether an ECG window is normal or abnormal by means of the AE-based anomaly detection workflow, reported in Fig. 3. The list of MSE errors obtained on the test dataset has been also used to define the anomaly detection threshold. This parameter is crucial since it estimates CAE performance on the task of anomaly detection, which identifies an anomalous window when it observes a reconstruction error above the threshold.

The workflow used to train and test our proposed CAE is depicted in Fig. 4, where the implemented CAE architecture is reported in Fig. 5. The model input consists of a 4-second signal window of 12 lead ECG sampled at 500 Hz.

The Encoder is composed of the following convolutional blocks:

- *Initial Encoder Block* - First two blocks (e.g., blocks 15 and 30 in Fig. 5) each consisting of Convolutional 2-dimensional layers, a Batch Normalization layer, Dropout and Tanh activation function;

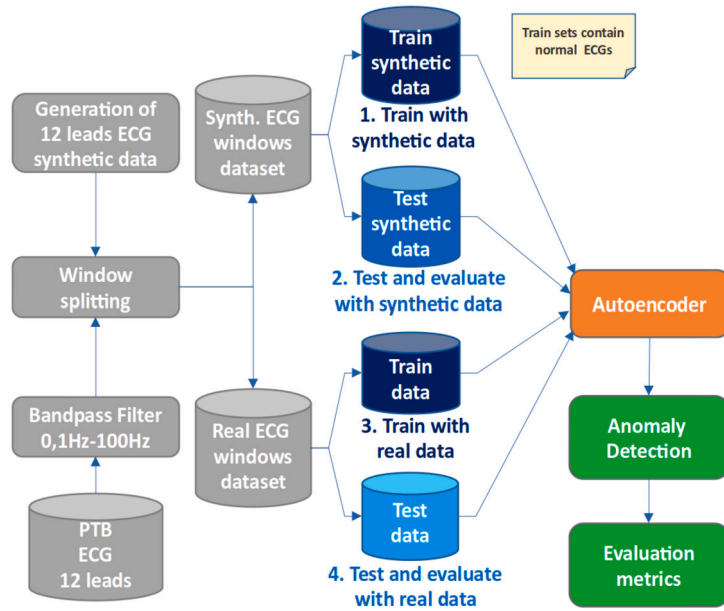


Fig. 4. Workflow used to train and test our proposed CAE. In the first two steps, only 12-lead synthetically generated ECG windows have been used. Then, transfer learning enables to retrain and evaluate our model in a real-world scenario with real ECG signals.

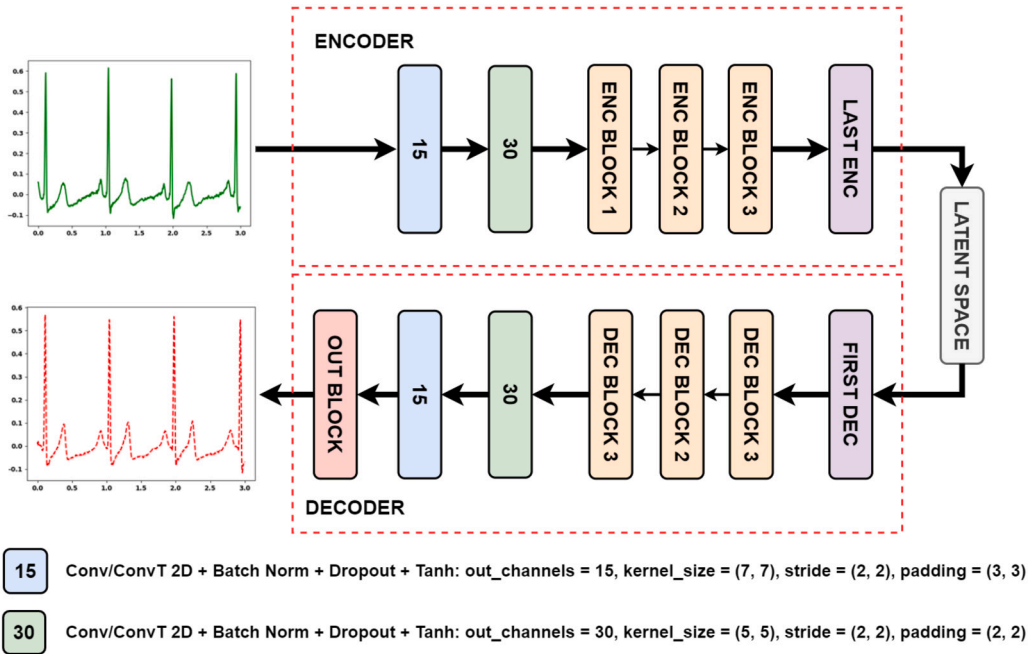


Fig. 5. Proposed CAE-based framework architecture. Note that input values (left side of the Figure) are represented by the 12 ECG leads. The encoded part is reported in the upper part of the figure, whereas the latent space signals are given as input to the decoding process.

- *Encoder Block* - Three blocks (e.g., ENC BLOCK 1, ENC BLOCK 2 and ENC BLOCK 3 in Fig. 5) each consisting of convolutional 2-dimensional layers, Tanh activation function, Batch Normalization layer, Dropout and skip connection layers;
- *Last Convolution Encoder Block* - Last block (e.g., LAST ENC block in Fig. 5) composed by Convolutional 2-dimensional layers and Tanh activation function.

Parameter values used for the encoder convolutional blocks are reported respectively in Table 2 and in Table 3. In both tables, the Convolutional 2-dimension layer, the Batch Normalization layer, Padding and Dropout are reported as Conv2D, BN2d, Pad and Do, respectively.

Table 2

Layers and parameters of the Encoder block in the Encoder module.

Layer	Input channel	Output channel	Kernel	Stride	Pad	Do
Conv2D+Tanh+BN2d	[15,30,30]	[15,30,30]	(3,3)	(1,1)	(1,1)	0.2
Conv2D	[15,30,30]	[30,30,45]	(3,3)	(2,2)	(1,1)	-
skipcon Conv2D + BN2d	[15,30,30]	[30,30,45]	(3,3)	(1,1)	(0,0)	-
BN2d + Tanh	[30,30,45]	[30,30,45]	-	-	-	0.2

Table 3

Layers and parameters of the Last convolutional Encoder block in the Encoder module.

Layer	Input channel	Output channel	Kernel	Stride
Conv2d + Tanh	60	70	(1,3)	(2,2)
Conv2d + Tanh	70	80	(1,3)	(2,2)
Conv2d + Tanh	80	90	(1,3)	(2,2)
Conv2d + Tanh	90	100	(1,3)	(2,2)
Conv2d + Tanh	100	128	(1,5)	(2,2)

Table 4

Layers and parameters of the Initial convolution Decoder block in the Decoder module.

Layer	Input channel	Output channel	Kernel	Stride	Out pad
ConvT2d+Tanh	128	100	(1,5)	(2,2)	(0,1)
ConvT2d + Tanh	100	90	(1,3)	(2,2)	(0,1)
ConvT2d + Tanh	90	80	(1,3)	(2,2)	(0,1)
ConvT2d + Tanh	80	70	(1,3)	(2,2)	(0,1)
ConvT2d + Tanh	70	60	(1,5)	(2,2)	(0,0)

Table 5

Decoder module: (latter) 1-dimensional convolutional block. Parameters and layer types.

Layer	Input channel	Output channel	Kernel	Stride	Pad	Groups
Conv1d + Tanh	15	15	(5,)	(1,)	(2,)	15
Conv1d + Tanh	15	15	(3,)	(1,)	(1,)	15
Conv1d + Tanh	15	15	(3,)	(1,)	(1,)	15
Conv1d + Tanh	15	15	(3,)	(1,)	(1,)	15

Table 6

The 2-dimensional convolutional transpose block used to construct the Decoder Module.

Layer	Input	Output	Kernel	Stride	Pad	Out pad	Do
ConvT2d + Tanh + BN2d	[45,30,30]	[30,30,15]	(3,3)	(1,1)	(1,1)	-	0.2
ConvT2d	[45,30,30]	[30,30,15]	(3,3)	(2,2)	(1,1)	(1,1)	-
skipcon ConvT2d + BN2d	[45,30,30]	[30,30,15]	(3,3)	(1,1)	(0,0)	(1,1)	-
BN2d + Tanh	45	45	-	-	-	-	0.2

The Decoder is composed of the following convolutional blocks:

- *Initial Convolution Decoder Block* - First block (e.g., FIRST DEC in Fig. 5) composed by Transposed Convolutional 2-dimensional layers and Tanh activation function;
- *Decoder Block* - Three blocks (e.g., DEC BLOCK 1, DEC BLOCK 2 and DEC BLOCK 3 in Fig. 5) each consisting of Transposed Convolutional 2-dimensional layers, Tanh activation function, Batch Normalization layer, Dropout and skip connection layers;
- *Final Decoder Blocks* - Two blocks (e.g., blocks 30 and 15 in Fig. 5) each consisting of a 2-dimensional Convolutional layer, a Batch Normalization layer, Dropout and Tanh activation function;
- *Output Convolutional Decoder Block* - Last block (e.g., OUT BLOCK in Fig. 5) composed by Convolutional 1-dimensional layers and Tanh activation function.

Layers and parameters used for the three decoder blocks reported above are summarized respectively in Table 4, Table 5 and Table 6. In these tables, the Transposed Convolutional 2-dimension layer, the Batch Normalization layer, Padding and Dropout are reported as ConvT2D, Pad and Do, respectively.

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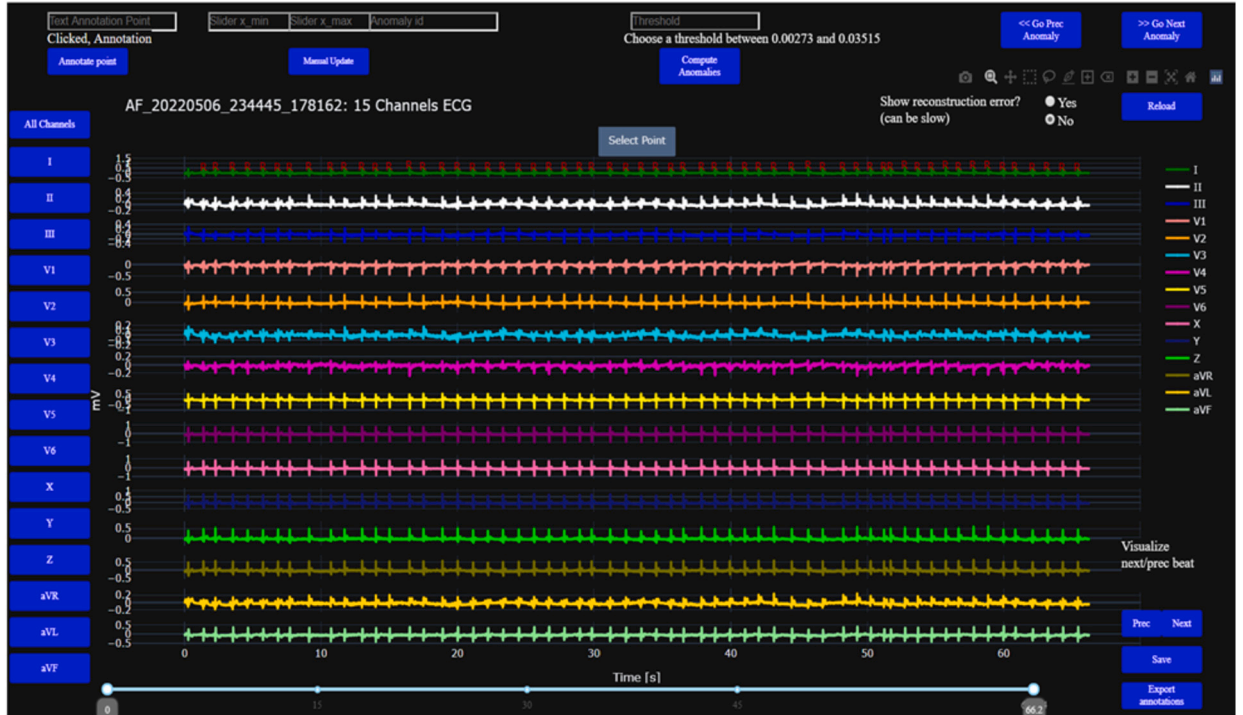


Fig. 6. GUI of the CAE-based framework for ECG anomalies detector. The interface allows also the user to annotate signals and export results.

The input channel defines the number of channels (or depth) of the input signal or feature map that the convolutional layer will process. The Output channel specifies the number of output channels generated by the convolutional layer. The Kernel defines the size of the filter applied during the convolution. Stride indicates the step size of the convolution across the input. A larger value for stride reduces the spatial resolution of the output. Padding adds a padding value to the input of the convolutional layer, which helps to control the output dimensions. Groups determine how the input and output channels are divided into groups. Output padding is used only in transposed convolution layers (i.e., layers inside the Decoder Blocks) to adjust the output dimensions.

Fig. 6 reports an example of the graphical user interface of the CAE-based framework. The physician may be able to visualize the anomalies that are identified by the module. The user may also annotate signals and the CAE may learn using the user annotations.

4. Experimental results

The CAE model has been trained and tested on both synthetic and real ECG data. Training has been performed using normal and noise-free synthetically generated ECG windows signals. Tests have been performed aiming to prove that CAE-based framework is able to discriminate between the synthetic ECG windows of normal and ECG with anomalies for instance presenting atrial fibrillation. The experimental tests have been performed also by using the MSE reconstruction error and identifying a threshold parameter useful for signal classification. In particular, the used workflow is the following Given an input 4-second window signal x and its reconstructed $\hat{x}$ returned from the CAE, we have that:

- If the MSE reconstruction error between the first 3 seconds of x and $\hat{x}$ is less than a given threshold parameter, then the window signal is considered normal. Despite that the AE takes in input 4 seconds, only three seconds have been used to compute the MSE error and predict the presence of an abnormality due to bad reconstruction of the last second present in the ECG window. E.g., Fig. 7 reports an example of a badly reconstructed signal in the last portion;
- in case of values over the threshold the input window signal is considered anomalous and presented to the clinician for final valuation and annotation.

Let TP and TN be the number of ECG windows correctly identified as normal or abnormal, respectively, and FP and FN be the number of normal or abnormal ECG windows misclassified, the model has been evaluated in terms of the following performance metrics:

$$\begin{aligned} \text{Accuracy} &= \frac{TP+TN}{TP+TN+FP+FN} \\ \text{Precision} &= \frac{TP}{TP+FP} \end{aligned}$$

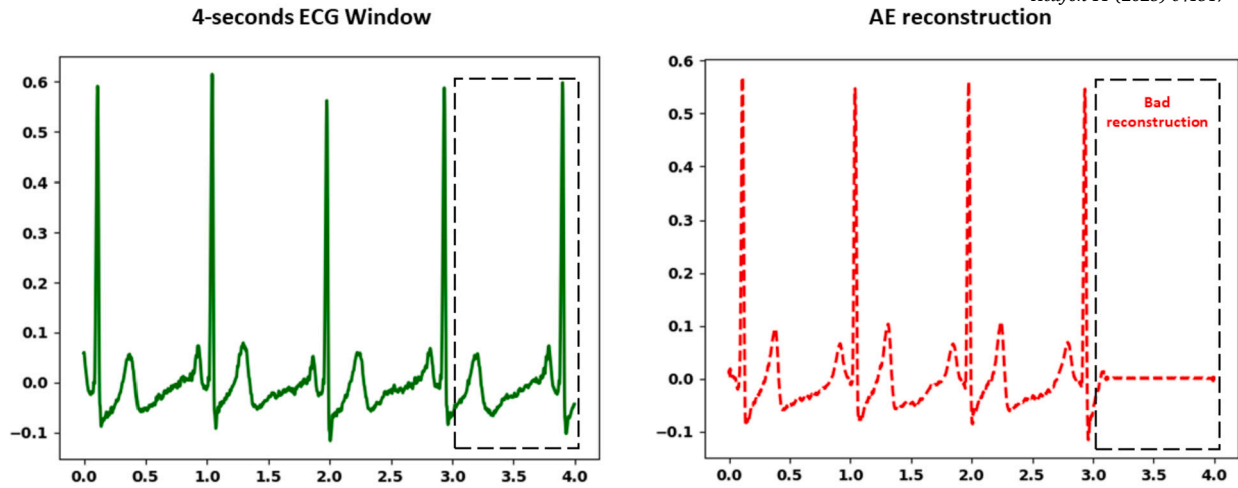


Fig. 7. Example of poor reconstruction in the last second of the signal.

Table 7

Performance metrics of our CAE in the anomaly detection task, using the synthetic dataset, with a threshold parameter equal to 0.0063.

Metric name	Metric value
Accuracy	97.19%
F1 score	94.84%
Sensitivity	99.14%
Specificity	96.50%
ROC AUC	99.72%
False Alarm Rate	0.86%
Missed Alarm Rate	3.5%

- $Sensitivity = Recall = \frac{TP}{TP+FN}$
- $F1 = \frac{2 * Precision * Recall}{Precision + Recall}$
- $Specificity = \frac{TN}{FP+TN}$
- $False\ Alarm\ Rate = \frac{FN}{TP+FN}$
- $Missed\ Alarm\ Rate = \frac{FP}{TN+FP}$
- ROC AUC is calculated as the Area Under the $Sensitivity - (1 - Specificity)$ curve, also known as the ROC curve.

The model was first evaluated and optimized regarding the AUC (Area Under Curve) ROC (Receiver Operating Characteristic) metric score on the synthetic test set. It represents the degree of separability between the two classes, i.e. normal and abnormal, providing an indicator of how well the model can distinguish between two classes. The model scored an AUC equal to 97.82% on the synthetic test set. Table 7 reports a number of classification performance metrics of our CAE in anomaly detection from the synthetic test set. Those metrics are obtained by optimizing the threshold parameter to maximize the ROC AUC score. The optimal threshold has been fixed equal to 0.0063.

The synthetic-trained model has been tested using all of the real ECG signals contained in the PTB dataset. A bandpass filter of 0.5 - 100 Hz was applied to remove noise, then a notch filter was applied to reject the 50 Hz noise from the electrical grid. After, the filtered signal was resampled from a sample rate of 1000 Hz to 500 Hz. Finally, a window splitting algorithm was applied to split the filtered signal in 4-second windows with a 1-second overlap.

To perform tests on the PTB dataset, it has been split into normal and signals with anomalies. Normal signals were split into 80% train and 20% test sets. The train set has been used to retrain the pre-trained model for 3000 epochs, with a batch size of 4 and an Adam optimizer with a 0.0001 learning rate. The final training set contains only healthy real ECG windows (some with noise). The final test set contains both healthy and unhealthy real ECG windows (some with noise). Train-test split details are reported in Table 8. In particular, the test data set is composed of 605 ECG healthy signals and 15377 signals presenting anomalies. The dataset consists of 4 second time windows for 12 leads ECG signals. Anomalies include the ones associated with pathologies as those reported in Table 1.

Fig. 8 shows two ROC curves obtained by using the CAE-based framework when the model is trained on the anomaly detection task and tested on the entire PTB dataset. It reports two cases: (i) when the model is trained on synthetic and free of noise ECG signals

Table 8
Train test split details.

Split	Normal windows	Abnormal windows
Train	2398	0
Test	605	15377

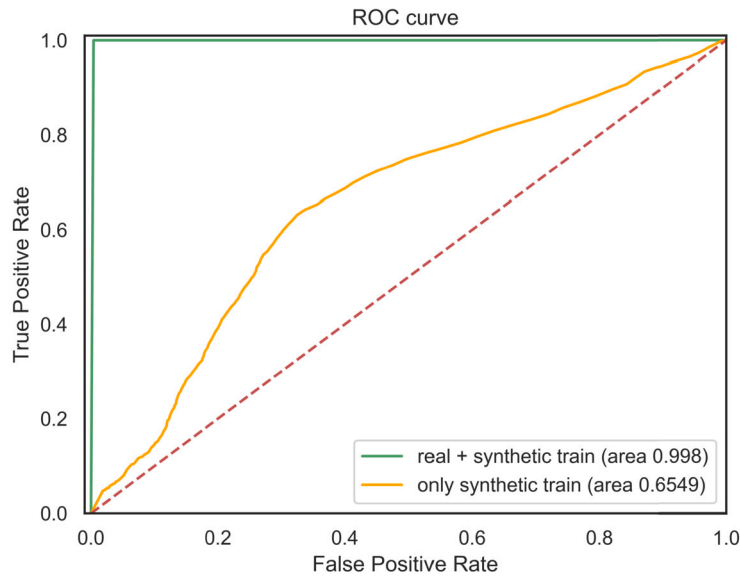


Fig. 8. ROC Curves obtained in the evaluation of our anomaly detection models over the entire PTB dataset.

Table 9
Performance metrics of the CAE model re-trained on the anomaly detection task, using the test set containing PTB real ECG signals and a threshold parameter equal to 0.0026.

Metric name	Metric value
Accuracy	99.97%
F1 score	99.59%
Sensitivity	99.50%
Specificity	99.99%
ROC AUC	99.75%
False Alarm Rate	0.5%
Missed Alarm Rate	0.01%

and (ii) when the model is retrained considering noise in real ECG signals. Table 9 reports the real-data classification performance metrics obtained by measuring the second case with the CAE model is retrained and tested on the anomaly detection task. These were obtained by optimizing the threshold parameter in order to maximize the ROC AUC score. The optimized threshold found is equal to 0.0026.

Table 10 reports classification metrics of the retrained CAE on the signal classification task using the optimized threshold of 0.0026 used above. Fig. 9 reports the ROC curve and the confusion matrix obtained on the signal classification task.

Tables 11 and 12 report results on the anomaly detection task compared with various state-of-the-art models carrying out anomaly detection on a real ECG dataset. Table 13 reports the performance of our proposed CAE in the detection of 12 different types of ECG abnormalities present in the PTB dataset.

The CAE-based framework has also been enriched by means of a graphical user interface presenting results and ECG signals. The GUI allows the user to load a large set of signals and read results on signal anomaly detection. Expert is also able to validate or add annotations to signals, where annotations are used as a human-in-the-loop protocol able to provide data useful to train the model. Thus the CAE-model can be trained using the user's additional annotations. Fig. 6 reports an example of an ECG leads report where users may add notes.

In particular, our methodology presents novelties with respect to existing applications of ECG autoencoders, in two main directions: (i) explainability and (ii) preprocessing techniques. In particular, regarding explainability, the tool allows to highlight the identified

Table 10

Performance metrics of our retrained CAE on the signal classification task, with the test set containing PTB real ECG signals and a threshold parameter equal to 0.0026.

Metric name	Metric value
Accuracy	99.56%
F1 score	93.33%
Sensitivity	87.50%
Specificity	100.00%
ROC AUC	93.75%
False Alarm Rate	12.50%
Missed Alarm Rate	0.0%

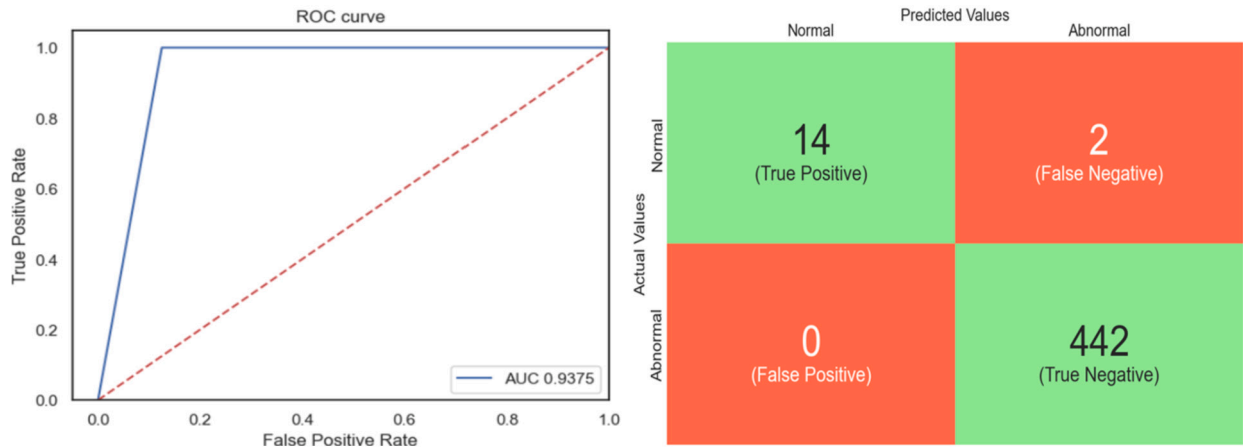


Fig. 9. ROC Curve (left) and Confusion matrix (right) obtained during the evaluation of our model on the signal classification task.

anomalies and characterize them with different meanings. Existing methods [35] employ global explainability by highlighting beats with high reconstruction errors, while the proposed method is able to differentiate whole areas of interest. In particular, the local explainability method proposed here is a *reconstruction error-based method* [25], which leverages signal reconstruction to explain abnormalities. This approach is particularly effective for autoencoder-based models. Building on the *reconstruction error-based method*, our approach reconstructs both normal and anomalous ECG patterns, producing lower reconstruction errors for normal patterns and higher reconstruction errors for anomalies. As illustrated in Fig. 10, the proposed tool presents the results to the physician as a colour-coded visualization, where the anomalies are highlighted using a coloured frame. This enables the system to both justify and localize the detected anomalies in the analyzed signal. Additionally, we evaluated the SHAP [27], [26] framework for explainability in anomaly detection. A comparative analysis of SHAP and our proposed approach indicates that explanations generated by SHAP are less reliable compared to those produced by our method. Visually, 10, the tool provides to the physician a coloured image where the implemented method provides the anomalies as a coloured frame. Thus, the system is able to justify and localize the proposed anomaly associated with the analyzed signal. We finally considered and tested the SHAP [27] approach for anomaly detection explainability. We compared SHAP results with those obtained by using the here proposed approach, and the explanations using SHAP are less reliable with respect to the ones obtained by using our method. CAE allows clinicians to test and verify the usability of the proposed methodology in a real data set to quickly identify anomalies associated with a large signal set. Tests have also been performed on real data samples.

The proposed method implements a local explainability approach that represents a novelty in relation to the existing methodology, thus supporting clinicians when deciding on ECG signal evaluation. Our methodology allows us to focus and drill down on the reconstructed signal by defining a small time interval, and to calculate the reconstruction error within that interval to locate abnormalities in the PQRST complexes. Fig. 10 highlights this difference by using different colours associated with areas and not with a single beat as in literature. Moreover, the coloured part is used to support clinicians in highlighting and explaining the detected anomalies in the signal patterns able to support clinicians in identifying and studying pathologies.

Finally, the CAE-based model is able to detect anomalies on 12 leads ECG signals and to support physicians in identifying from a large volume of data signals anomalies related to pathologies such as arrhythmia, atrial fibrillation, myocardial infarction, and can be used as a freely available tool (see GitHub repository https://github.com/UgoLomoio/ECG_DSS_CAE).

To validate our explainability approach, we randomly selected a subset of anomalous ECG windows from the PTB dataset, comprising 50 clinical cases. These ECGs were analyzed by a cardiologist who provided both diagnoses and annotations of abnormalities. Using these clinical annotations as ground truth, we validated the explainability results of our method by computing a quantitative metric to measure the alignment between our annotations and those of the clinician. The accuracy is evaluated by comparing the

Table 11

Comparison of the proposed model with related works in terms of performance (in bold the highest scores). Abbreviations: ESTT stands for European ST-T dataset; MB stands for MIT-BIH; Nor. stands for Normal number of signals, Abn. stands for the number of signals presenting anomalies. In the Model column, the bibliography references have been reported. The here proposed model is called CAE-based. The p-value is evaluated by using z-score and it is present when evaluation was possible.

Model	Dataset	Nor./Abn.	AUC	Sens	Spec	Acc	F1	p-value
[40]	MIT-BIH (2-leads)	143/713	97.06	99.02	93.01	98.01	0.988	-
[41]	MIT-BIH	902/20807	0.48	92.08	93.13	92.13	95.73	-
[41]	MIT-BIH	361/8322	0.95	99.16	43.88	94.90	0.973	5.029e-05
[42]	MIT-BIH	1005/1000	0.79	80.50	-	80.40	0.785	-
[43]	MIT-BIH	1005/1000	0.89	91.00	-	89.10	0.927	-
[44]	MIT-BIH	1005/1000	0.94	89.50	-	90.40	0.93	-
[45]	MIT-BIH	1005/1000	0.94	90.70	-	92.90	0.955	-
[46]	MIT-BIH	1005/1000	0.96	91.80	-	94.30	0.95	-
[47]	MB+ESTT	2653/2653	0.9974	99.74	99.85	99.79	0.9979	6.69e-17
CAE-based	PTB 12-leads anomaly detection	605/15377	0.9974	99.50	99.99	99.97	0.996	0.0024
[48]	PTB myocardial infraction detection	-	-	99.40	99.45	99.43	-	-
CAE-based	PTB myocardial infraction detection	605/13163	0.9975	99.50	100	99.98	0.9975	-
[16] Only QRS atrial fibrillation	PTB-XL (train) CPSC 2018 (test)	6460/1347	0.74	56.00	-	56.00	0.56	-
CAE-based Only QRS atrial fibrillation	PTB-XL (train) CPSC 2018 (test) 12-leads	3719/4267	0.80	82.47	77.83	80.00	0.79	-

Table 12

Comparison among the proposed model and the existing ones with respect to False Alarm Rate (FAR), Missed Alarm Rate (MAR), dataset, and method.

Model	Dataset	Nor./Abn.	FAR	MAR	p-value
[40]	MIT-BIH	143/713	0.98%	6.99%	-
[41]	MIT-BIH	902/20807	7.91%	6.87%	-
[41]	MIT-BIH	361/8322	0.84%	56.12%	0.8070
[47]	MIT-BIH + European ST-T	2653/2653	0.26%	0.15%	0.0008
CAE-based	PTB	605/15377	0.5%	0.01%	0.0482

Table 13

Classification metrics of the proposed model in the detection of different types of abnormalities, with a fixed and optimized reconstruction error threshold of 0.0026.

Class	Nor./Abn.	AUC %	Sens. %	Spec. %	Acc %	F1%	FAR %	MAR %
All	605/15377	99.74	100	99.50	99.97	99.59	0.5	0.01
Bundle branch block	605/607	99.75	99.50	100	99.75	99.75	0.5	0.0
Cardiomyopathy	605/587	99.75	99.50	100	99.75	99.75	0.5	0.0
Dysrhythmia	605/393	99.75	99.50	100	99.70	99.75	0.5	0.0
Heart failure	605/30	99.75	99.50	100	99.51	99.75	0.5	0.0
Hypertrophy	605/265	99.56	99.50	99.62	99.54	99.67	0.5	0.038
Myocardial infarction	605/13163	99.75	99.50	100	99.98	99.75	0.5	0.0
Myocarditis	605/152	99.42	99.50	99.34	99.47	99.67	0.5	0.66
Palpitation	605/10	99.75	99.50	100	99.51	99.75	0.5	0.0
Stable angina	605/47	99.75	99.50	100	99.54	99.75	0.5	0.0
Unstable angina	605/10	99.75	99.50	100	99.51	99.75	0.5	0.0
Valvular heart disease	605/113	99.75	99.50	100	99.58	99.75	0.5	0.0

Patient: patient150, Window: 4, Reconstruction error: 0.0049, Target: Normal, Prediction: Anomalous

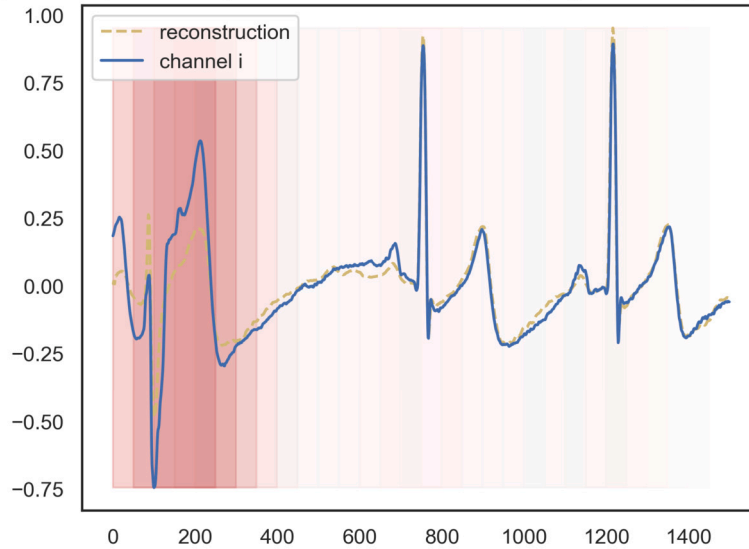


Fig. 10. An example of explainability results using our framework on a healthy ECG window with structural anomalies not related to cardiac diseases.

(a) Anomalies reported manually by cardiologist

(b) Explanations returned by CAE as coloured part of the signals

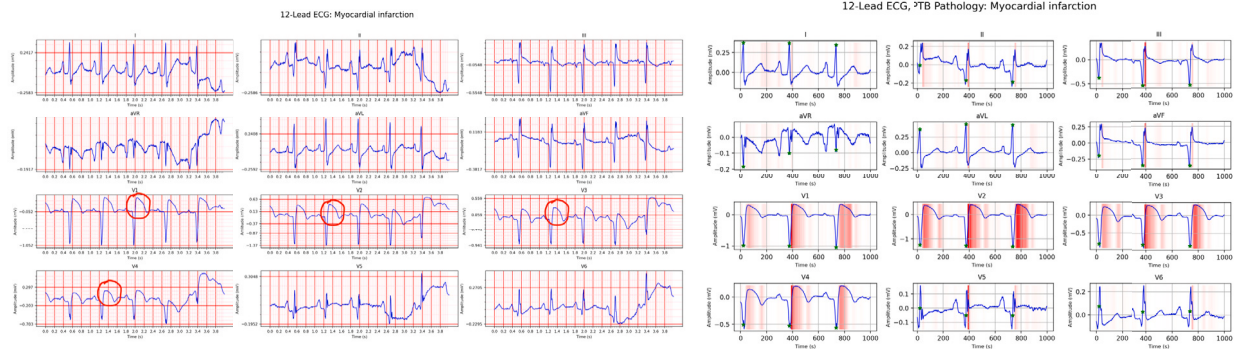


Fig. 11. Figure reports a subset of ECG windows with myocardial infarction extracted from PTB dataset where anomalies have been annotated in (a) by a specialist and in (b) annotated by the explanation module of the here proposed method. On the right part of the figure, the coloured part corresponds to the annotations reported by physician proving the reliability of the proposed model.

number of correctly identified anomalies versus the total number of annotated signal windows. Our approach achieved an accuracy of 97.78% in detecting and explaining abnormalities within ECG signals, thereby validating the effectiveness of our explainability method. Fig. 11 illustrates an example of an ECG window annotated by the clinician (left side of the Figure) and by our model (right side of the Figure).

5. Discussion and limitations

The CAE was initially trained using synthetic, noise-free normal ECG windows from the PTB dataset. The synthetic signals were generated using the SIMARR3 software, and the CAE achieved an AUC score of 97.82% on anomaly detection tasks. To minimize the effects of noise in the PTB dataset, the model was retrained using 80% of the normal signals. The remaining 20% of normal signals, along with all abnormal signals, were used to create the test set.

The performance of the retrained CAE was evaluated on the anomaly detection tasks using this test set, achieving an AUC score of 99.74% and a MAR of 0.01%. Our model performs on par with, and in some classification metrics even surpasses, the state-of-the-art model for anomaly detection in ECG signals. The experiments demonstrate that the model performs efficiently, making very few errors, which can be attributed to the quality of the PTB dataset, where there is a clear distinction between normal and noisy ECG signals, and only a few errors are made in classifying anomalous windows.

We performed a comparative test with respect to recent publications and with a slightly different dataset. In particular, we used the PTB-XL dataset [32], and we compared the performance with the experiments reported in [16]. We trained our CAE-based framework

by using the PTB-XL dataset and we performed tests on the China Physiological Signal Challenge 2018 (CPSC-2018 [33]) reproducing one of the experiments proposed in [16]. The AUC ROC results indicate that the here proposed framework performs better in detecting atrial fibrillation inside the QRS complex (AUC ROC = 0.8 versus AUC ROC = 0.74 of [16]).

Since no other studies in literature perform general anomaly detection on the PTB dataset, to compare our model trained on the PTB we used our model to detect myocardial infarction and then compared performances against the one reported in [48].

In summary, the CAE-based tool yields few false negatives, meaning it rarely classifies normal windows as abnormal. While this could be considered a potential limitation, it is largely due to the nature of the dataset used. The tool proves to be a valuable asset for clinicians in detecting (even hidden) anomalies in ECG signals, which may be associated with clinically significant pathologies. Although we believe that diagnoses should ultimately be made by physicians, incorporating annotations into the dataset may further enhance the CAE-based framework, transforming it into a powerful neural network tool to support clinical decision-making. Additionally, the tool can reconstruct ECG signals with high precision and efficiency, and it imposes no limitations on the duration of ECG recordings due to its window-splitting approach. However, a limitation arises when dealing with highly noisy ECG signals, where the CAE framework struggles to reliably detect anomalies related to cardiological events. The here proposed CAE architecture is optimized to perform anomaly detection in 12-leads ECG datasets, this limits our comparison against state-of-the-art models. For example, comparing our model's performance to the literature ones trained on the MIT-BIH dataset (only 2 leads) would not provide a fair evaluation.

Our proposed method implements a local explainability approach representing a novelty with respect to the existing methodologies in 12-lead ECG anomaly detection [35], aiming to give clinicians more detailed insights on abnormalities location (i.e. abnormal P wave). The here proposed local explainability method is a reconstruction error based method that uses signal reconstruction to explain abnormalities. The tool provides to the physician an image where anomalous patterns are red coloured, with intensity of the colour proportional to the reconstruction error. Explainability results were validated using abnormalities annotated by a clinical expert as ground truth. During the whole annotation task, clinician is blind about our model explanations or PTB signal diagnosis. We report 97.78% of accuracy in detecting and explaining local abnormalities in a subset of 50 randomly selected ECG signals. The proposed CAE tool employs a deep learning model optimized for 12-lead ECG signals. The primary limitation of this approach is that anomaly detection is restricted to ECG signals with 12 leads. Nevertheless, in its current form, the tool also functions as annotation software, allowing for the annotation and learning from ECG signals with 12 leads. Finally, we assert that even in its current version, the CAE-based framework can serve as a valuable support tool for physicians in making diagnostic decisions.

6. Conclusions and future works

The presented CAE-based framework can be used to automatically identify anomalies in ECG signals. It allows automatically identifying anomalies that can be correlated to atrial fibrillation events, myocardial infarction, or noise disturbance in the analyzed ECG signal. The tool is freely available on a GitHub repository and can be used to enrich and produce additional annotated ECG signals useful to retrain the model.

Future works will focus on using a CAE-based framework in a cloud-based environment where clinicians may annotate and enrich datasets and where signals can be analyzed in a remote control mechanism. Thus, the CAE-based framework may allow the possibility of sharing signals and learning from a collaborative environment. Finally, the framework can be used in any context when signals to be analyzed present a long acquisition time.

Summary point

ECG signals; Autoencoder; Arrhythmia; Classification; denoising; support system.

CRedit authorship contribution statement

Ugo Lomoio: Software, Data curation. **Patrizia Vizza:** Writing – original draft, Methodology, Conceptualization. **Raffaele Giancotti:** Visualization, Software, Data curation. **Salvatore Petrolo:** Software. **Sergio Flesca:** Methodology, Investigation, Conceptualization. **Fabiola Boccuto:** Methodology, Formal analysis. **Pietro Hiram Guzzi:** Writing – review & editing, Writing – original draft, Validation, Supervision. **Pierangelo Veltri:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Giuseppe Tradigo:** Writing – original draft, Supervision, Conceptualization.

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Declaration of competing interest

We here declare that none of the coauthors have competing interests.

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Availability of data and materials

Pretrained models, setup, code, and user guide presented or used in this work are available at: https://github.com/UgoLomoio/ECG_DSS_CAE.

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